(Please write your Enrollment Number) Enrollment No. 01601172021

MID-TERM EXAMINATION
(Course Name – B. Tech, CSE-AI, ECE-AI) (Semester - 4)
(February, 2023) OFFLINE mode

Subject Code: BAI 204	Subject: Optimization Technique and Decision Making

Time : 1 ½Hours Maximum Marks : 30

Note: Q. 1 is compulsory.

Q1 (2.5*4)

(a) Discuss the statement of an optimization problem.

(b) Describe objective function and constraints with the help of an example.

(c) Differentiate between linear programming problems and quadratic programming problems

(d) What is optimization? Explain with an example.

Q2 **(Attempt any Two Parts)** (5,5)

(a) Discuss the general structure of the optimization problem with the help of a flowchart.

(b) What are the various ways to classify optimization problems?

(c) Discuss five applications of optimization in engineering.

Q3 **(Attempt any Two Parts)** (5,5)

(a) Discuss integer programming using an example.

(b) A factory produces 2 products p1 and p2. Each of the products p1 requires 2 hrs for molding, 3 hrs for grinding, and 4 hrs for polishing, and each of the products p2 requires 4 hrs for molding, 2 hrs for grinding, and 2 hrs for polishing. The factory has molding machine available for 20 hrs, grinding machine available for 24 hrs, and polishing machine available for 13 hrs. the profit is Rs. 5 per unit of p1 and Rs. 3 per unit of p2, and the factory can sell all that it produces. Formulate the problem as a linear programming problem to maximize profit. Does an optimal solution exist for maximization? If yes, state the values of design variables as well as the optimal solution.

(c) Solve the following problem using the Simplex method

Max $Z=5X_1+3X_2$

Subject to constraints:-

$X_1 + X_2 <= 2$

$5X_1 + 2X_2 <= 10$

$3X_1 + 8X_2 <= 12$

$X_1 , X_2 >= 0$